

Alassane Diop

ML / AI Engineer

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PROFILE

ML/AI Engineer with 5 years of experience building and shipping AI systems — recommendation engines, computer vision applications, LLM pipelines, and scalable data infrastructure. Proven track record taking models from research to production on GCP, including Mitacs-funded rec system work deployed live. Comfortable owning the full ML lifecycle: data, modeling, inference, monitoring.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript

AI / ML: LLMs (Claude, OpenAI, Gemini), RAG, Recommendation Systems, Computer Vision, NLP, Embeddings, Similarity Search, PyTorch, scikit-learn, Finetuning

Data & Infra: Snowflake, dbt, Snowflake Cortex, GCP, REST APIs, Data Pipelines, Semantic Layer Design, Data Modeling

Engineering: React Native (Expo), Next.js, macOS App Development, Git, CI/CD

EXPERIENCE

Data Scientist — ML Engineering

July 2023 – Present

Browns Shoes — Montréal, Canada

- Built a product recommendation engine generating personalized suggestions across the full e-commerce catalog
- Developed a shoe recognition app using computer vision to identify products from user photos and return matching catalog entries with full product details
- Designed and built an NPS AI pipeline for automated sentiment analysis and insight extraction from customer feedback using LLMs
- Other notable initiatives: AISQL-to-Snowflake Cortex REST API migration with semantic layer design, multi-project dbt architecture with RBAC, and an attribution model linking online traffic to in-store purchases

ML Engineer / Founder

2025 – 2026

Olima.ai & Vibecope.ai — Montréal, Canada

- Built and launched Olima.ai, a mobile game (Expo/React Native) powered by a celebrity similarity algorithm using Wikipedia clickstream data, spectral embeddings, and a custom recommender system
- Built and launched Vibecope.ai, a macOS native menu bar app that coaches AI prompting in real-time via a UserPromptSubmit event hook; shipped with Apple Developer ID signing, notarized DMG distribution

Analyst, CRM & Customer Analytics

Mar 2022 – July 2023

ALDO Group — Montréal, Canada

- Built predictive models for segmentation, churn, and lifetime value; designed scoring systems and market basket analysis to optimize personalized marketing
- Prototyped an AI-powered QA tool (GPT + HuggingFace) to detect misleading product color descriptions on the e-commerce platform, reducing return risk

AI Engineer (Consultant)

June 2021 – June 2023

Tastet.ca — Montréal, Canada

- Designed and deployed a hybrid recommender system on GCP, benchmarking 8 models (KNN, ALS, SVD, RankFM, cosine similarity, etc.) against both F1-score and a custom fairness metric to balance accuracy with merchant equity
- Architected event-driven ML pipeline with GCP Cloud Functions triggered on user/merchant registration, and weekly cron-based retraining/refresh to keep recommendations current
- Took the system from supervised research to live production; secured multiple rounds of Mitacs funding and authored a research thesis on state-of-the-art recommender systems and ethical AI

Data Engineer (Intern)

Sept 2018 – Sept 2020

ADNCorp — Dakar, Sénégal

- Data modeling, quality assessment, and R&D prototyping; master's thesis on production monitoring with error tracking for a large-scale manufacturer

EDUCATION

M.Sc. Data Science & Business Analytics

December 2021

HEC Montréal — GPA: 3.77 / 4.3

Computer Engineering Degree (M.Sc. equivalent)

September 2020

École Polytechnique de Thiès, Senegal — Top of class — Thesis defense: 18.8/20